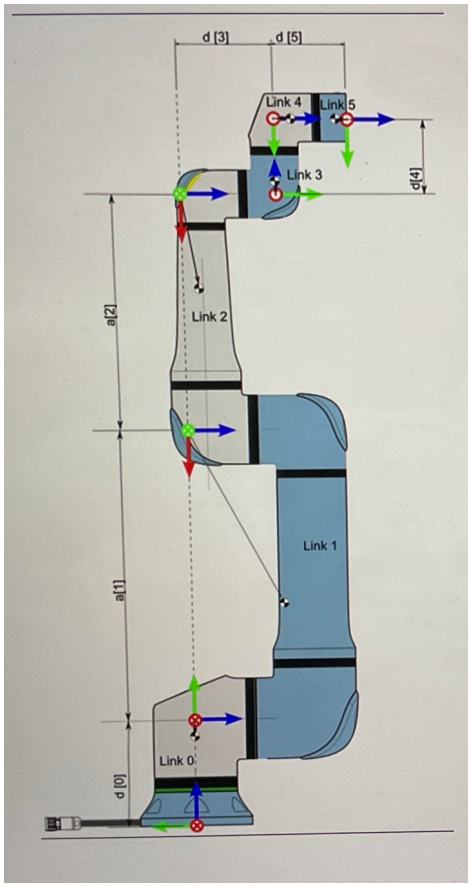
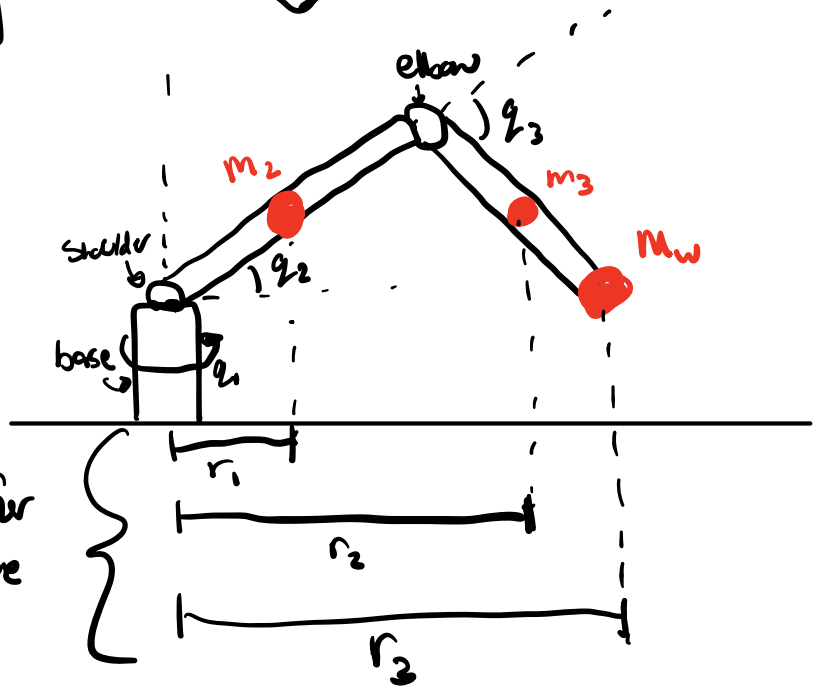


* URSe robot arm
* Take first 3 joints (base, shoulder, elbow) and freeze 3 wrist joints
* link inertia masses ($\underline{I}$) are negligible as specified in specs
  - treat each link as point mass

Quantity	Value	Significance
joint range (base, shoulder, elbow)	$\pm 360° = \pm 6.283$ rad	position limits
max joint speed	$\pm 180° = \pm 3.142 \frac{rad}{s}$	speed limits
max joint torque	120 N·m	torque limits
max joint acceleration	1.4 rad/s^2	acceleration limits
upper arm length L_2	0.425 m	kinematics
fore arm length L_3	0.392 m	kinematics
Shoulder height	0.1625 m	kinematics
Upper arm mass / COM x	8.058 kg / 0.2125 m	torque calc

forearm mass / COM x	2.846 kg / 0.15 m	torque calc
wrist links 4,5,6	1.37+1.3+0.365 = 3.035 kg	load on forearm tip

$$\underline{q}_0 = \begin{bmatrix} 0 \\ 0.6 \\ -1.2 \end{bmatrix}$$

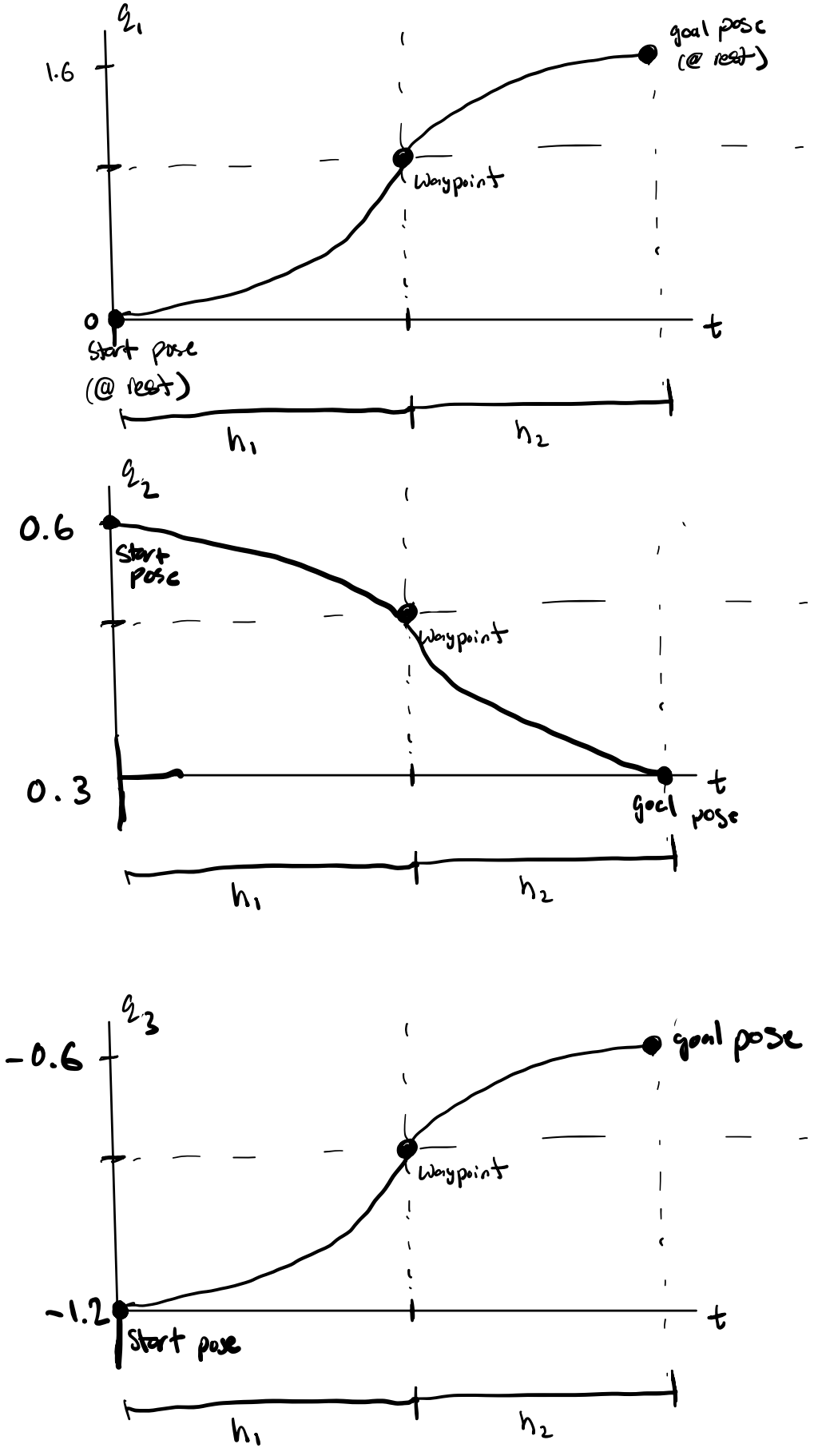


Used for Torque calc

$$\underline{q}_0^{(0)} = \begin{bmatrix} 0 \\ 0.6 \\ -1.2 \end{bmatrix} \Longrightarrow \underline{q}_g^{(2)} = \begin{bmatrix} 1.6 \\ 0.3 \\ -0.6 \end{bmatrix}$$

$$\Delta \underline{q} = \begin{bmatrix} 1.6 \\ -0.3 \\ 0.6 \end{bmatrix}$$

Calc reach to make sure it is less than or equal to max reach


q_1
1.6
goal pose
(@ rest)
waypoint
0
t
start pose
(@ rest)
h_1
h_2
q_2
0.6
start pose
waypoint
0.3
t
goal pose
h_1
h_2
q_3
-0.6
goal pose
waypoint
-1.2
t
start pose
h_1
h_2

$$\min \quad f(\underline{x}) = h_1 + h_2$$

$$\text{w.r.t} \quad \underline{x} = [q_1, q_2, q_3, h_1, h_2] \in \mathbb{R}^5$$

$$\text{S.T} \quad g_1(\underline{x}) = q_{i,K}(\underline{x}) - 6.283 \leq 0$$

$$g_2(\underline{x}) = -q_{i,K}(\underline{x}) - 6.283 \leq 0$$

$$g_3(\underline{x}) = \dot{q}_{i,K}(\underline{x}) - 3.142 \leq 0$$

$$g_4(\underline{x}) = -\dot{q}_{i,K}(\underline{x}) - 3.142 \leq 0$$

$$g_5(\underline{x}) = \ddot{q}_{i,K}(\underline{x}) - 1.4 \leq 0$$

$$g_6(\underline{x}) = -\ddot{q}_{i,K}(\underline{x}) - 1.4 \leq 0$$

$$g_7(\underline{x}) = \tau_i(\underline{x}) - 120 \leq 0$$

$$g_8(\underline{x}) = -\tau_i(\underline{x}) - 120 \leq 0$$

$$g_9(\underline{x}) = 6d_{i,K}(\underline{x}) - j_{max} \leq 0$$

$$g_{10}(\underline{x}) = -6d_{i,K}(\underline{x}) - j_{max} \leq 0$$

need to decide (true as well)

$$g_{11}(\underline{x}) = 0.1 - h_K \leq 0$$

$$g_{12}(\underline{x}) = h_K - 5 \leq 0$$

segment duration

$$0.1 \leq h_K \leq 5$$

Torque

$$\underline{\tau} = \underbrace{\underline{M}(\underline{q})\ddot{\underline{q}}}_{\text{to accelerate}} + \underbrace{\underline{C}(\underline{q},\dot{\underline{q}})\dot{\underline{q}}}_{\text{centrifugal \& coriolis}} + \underbrace{\underline{G}(\underline{q})}_{\text{gravity}}$$

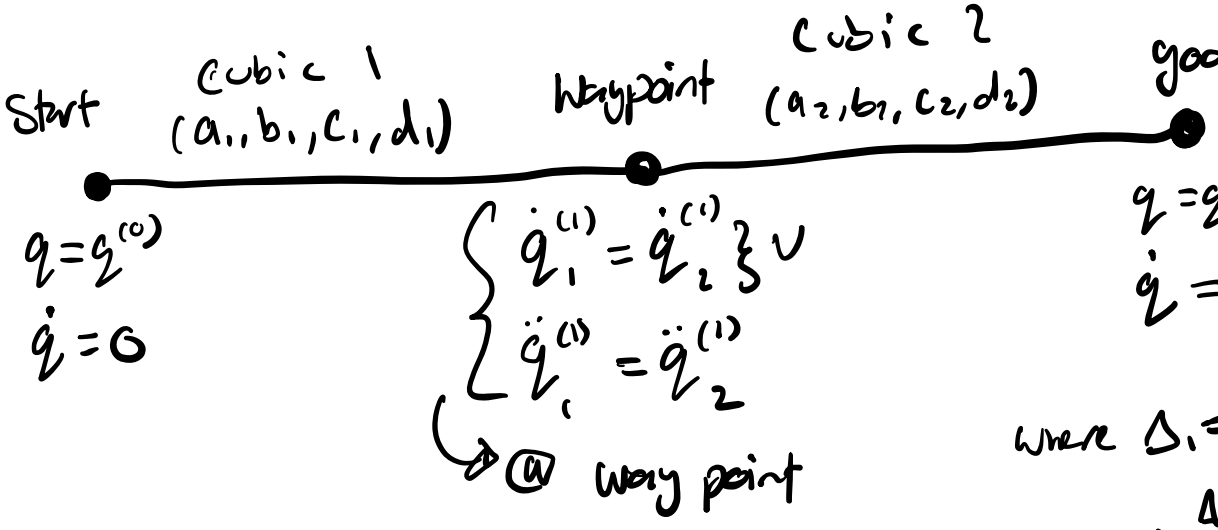
Cubic Spline (assumption)

$$q(s) = a + bs + cs^2 + ds^3, \quad s = \text{time since segment started}$$

$$\dot{q}(s) = b + 2cs + 3ds^2$$

$$\ddot{q}(s) = 2c + 6ds$$

jerk $\Rightarrow \dddot{q}(s) = 6d$



where $\Delta_1 =$ waypoint pose − start pose

$\Delta_2 =$ goal pose − waypoint pose

let $\dot{q}_1^{(1)} = \dot{q}_2^{(1)} = v = \dfrac{1.5\left(\Delta_1/h_1^2 + \Delta_2/h_2^2\right)}{1/h_1 + 1/h_2}$

then we solve for a, b, c, d

then we plug s (time)

then we calculate for $q, \dot{q}, \ddot{q}, \dddot{q}$ (jerk)

then we calculate for T

add all vals to constraint, then run solver